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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-2024

Program: BTECH

Year: III Semester: VI

Stream : Cyber Security (CSECS)

Subject: Digital Forensics and Incident Response

Time: 3 hrs (2:00pm to 5:00pm)

Date: 26/ 6/2024

No. of Pages : 2

Marks: 100

Re Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO-1 ; SO-2 ; BL-2	a.	What is the role of an expert witness in legal proceedings, and how do they contribute to the judicial process?	5
CO-1 ; SO-2 ; BL-2	b.	What is the difference between watermarking and steganography and what are they typically used for?	5
CO-1 ; SO-2 ; BL-3	c.	A business impact analysis indicates an organization cannot operate without its web server for more than 4 days and still recover. The mean time to repair is 1 day. How many days do you have after a disaster to initiate repair or will the organization not be able to recover?	5
CO- 1; SO- 2; BL-2	d.	What are sectors in the context of computer storage, and how do they contribute to data organization and retrieval?	5
Q2 CO-1 ; SO-2 ; BL-3	a	In mobile forensics, what are some of the direct types of evidence that you can collect and what are the indirect types of evidence? Which one of these can be a direct type of evidence in cyberstalking cases but indirect in other cases? Also, list the locations inside a mobile device where information might be stored that should be retrieved for when doing a forensics investigation.	10
CO-3 ; SO-2 ; BL-2	b	i) Explain your understanding about Cyber law in India ? [5] ii) Differentiate Between Morals, Ethics, and Laws? [5]	10

Q3 CO-1; SO2-; BL-2	a	You are investigating a cyberattack that infiltrated a corporate network through a compromised router. The network security team has preserved a forensic image of the router's flash memory. Describe the steps you would take to analyze the router's flash memory for evidence of the cyberattack.	10
CO-1; SO-2; BL-3	b	The Tor browser offers a level of anonymity online, but it's not without limitations. Explain how the Tor network anonymizes user traffic. Discuss two potential security vulnerabilities associated with using the Tor browser, and describe how a user can mitigate these vulnerabilities.	10
Q4 CO-1; SO-2; BL-2	a	NTFS (New Technology File System) is the primary file system used by modern Windows operating systems. Explain the following aspects of NTFS: • Describe two key advantages of NTFS compared to older file systems like FAT32. • Explain the concept of journaling in NTFS and its benefit for data recovery. • Discuss two security features offered by NTFS that enhance data protection. • Briefly mention one potential drawback of using NTFS compared to simpler file systems	10
CO-3; SO-2; BL-2	b	A critical component of incident response (IR) is prioritizing vulnerabilities and taking appropriate actions. Explain how the Common Vulnerability Scoring System (CVSS) can be used in an IR process, considering both its strengths and weaknesses.	10
Q5 CO-1; SO-2; BL-3	a	During a forensic investigation, you recover a password file containing username and hash values. However, you don't have the original passwords. Explain two common password hashing techniques and discuss their relative strengths and weaknesses in terms of security and password cracking.	10
CO-1; SO-2; BL-3	b	Several graphic files (e.g., JPEG, PNG) were received via email from an unknown source by a suspect in an investigation. You suspect these files might contain hidden messages. Describe the procedures you would follow to extract any hidden data and what tools you might use.	10
Q6 CO-1; SO-2; BL-2	a	Describe the structure of a report – what are the various sections and what should they contain? What are some of the factors to keep in mind to make the reports clear and precise?	10
CO-1; SO-2; BL-2	b	Denial-of-Service (DoS) attacks can cripple online services. Explain two preventative measures and two reactive measures that can be taken to mitigate DoS attacks.	10
Q7 CO-2; SO-2; BL-2	a	A critical aspect of cybersecurity is a well-defined incident response process. Describe the five main stages of an incident response process and briefly explain the purpose of each stage.	10
Q7 CO-3; SO-2; BL-2	b	Digital forensics plays a crucial role in legal proceedings, but it also faces unique challenges. Describe two legal challenges that can arise when presenting digital evidence in court, and explain how these challenges can be mitigated.	10